Stock Photo

3000 BC - 1776	1911 - 1947	Late 1700s - 1950s	1940s - 1950s	1960s - present
Early Management	Classical Approach	Behavioral Approach	Quantitative Approach	Contemporary Approaches

CLASSICAL Approach

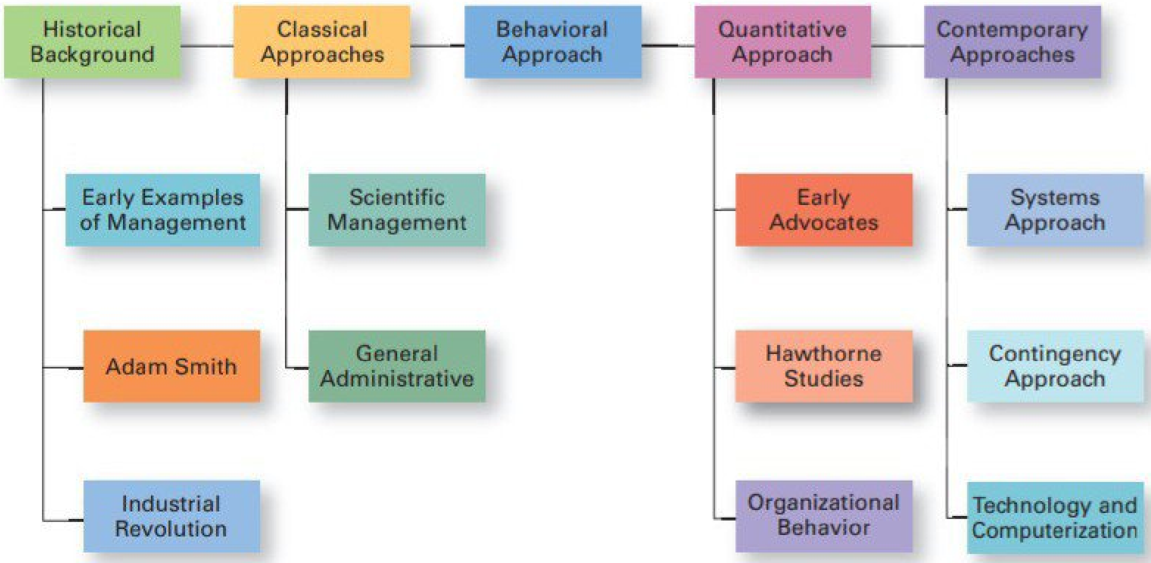
MH1.2

Although we've seen how management has been used in organized efforts since early history, the formal study of management didn't begin until early in the twentieth century. These first studies of management, often called the **classical approach**, emphasized rationality and making organizations and workers as efficient as possible. Two major theories compose the classical approach: scientific management and general administrative theory. The two most important contributors to scientific management theory were Frederick W. Taylor and the husband-wife team of Frank and Lillian Gilbreth. The two most important contributors to general administrative theory were Henri Fayol and Max Weber. Let's take a look at each of these important figures in management history.

Scientific Management

If you had to pinpoint when modern management theory was born, 1911 might be a good choice. That was when Frederick W. Taylor's *Principles of Scientific Management*

Exhibit MH-1
Major Approaches to Management



Frederick Taylor
Frederick Taylor's *Principles of Scientific Management*, published in 1911, described his theory of scientific management.
Source: Jacques Boyer/Roger-Viollet/The Image Works

was published. Its contents were widely embraced by managers around the world. Taylor's book described the theory of **scientific management**: the use of scientific methods to define the "one best way" for a job to be done.

Taylor worked at the Midvale and Bethlehem steel companies in Pennsylvania. As a mechanical engineer with a Quaker and Puritan background, he was continually appalled by workers' inefficiencies. Employees used vastly different techniques to do the same job. Virtually no work standards existed, and workers were placed in jobs with little or no concern for matching their abilities and aptitudes with the tasks they were required to do. They often "took it easy" on the job, and Taylor believed that worker output was only about one-third of what was possible. Taylor's solution was to apply the scientific method to shop-floor jobs. He spent more than two decades passionately seeking the "one best way" for each job to be done.

Taylor's experiences at Midvale led him to define clear guidelines for improving production efficiency. He argued that these four principles of management (see Exhibit MH-2) would result in prosperity for both workers and managers.[3] How did these scientific principles really work? Let's look at an example.

Probably the best-known example of Taylor's scientific management efforts was his pig iron experiment. Workers loaded "pigs" of iron (each weighing 92 pounds) onto rail cars. Their daily average output was 12.5 tons. However, Taylor believed that by scientifically analyzing the job to determine the "one best way" to load pig iron, output could be increased to 47 or 48 tons per day. After scientifically applying different combinations of procedures, techniques, and tools, Taylor succeeded in getting that level of productivity. How? By putting the right person on the job with the correct tools and equipment, having the worker follow his instructions exactly, and motivating the worker with an economic incentive of a significantly higher daily wage. Using similar approaches for other jobs, Taylor was able to define the "one best way" for doing each job. Overall, Taylor achieved consistent productivity improvements in the range of 200 percent or more. Based on his groundbreaking studies of manual work using scientific principles, Taylor became known as the "father" of scientific management. His ideas spread in the United States and to other countries and inspired others to study and develop methods of scientific management. His most prominent followers were Frank and Lillian Gilbreth.

industrial revolution
A period during the late eighteenth century when machine power was substituted for human power, making it more economical to manufacture goods in factories than at home

classical approach
First studies of management, which emphasized rationality and making organizations and workers as efficient as possible

scientific management
An approach that involves using the scientific method to find the "one best way" for a job to be done

Exhibit MH-2

Taylor's Scientific Management Principles

1. Develop a science for each element of an individual's work to replace the old rule-of-thumb method.
2. Scientifically select and then train, teach, and develop the worker.
3. Heartily cooperate with the workers to ensure that all work is done in accordance with the principles of the science that has been developed.
4. Divide work and responsibility almost equally between management and workers. Management does all work for which it is better suited than the workers.

Source: F. W. Taylor, *Principles of Scientific Management* (New York: Harper, 1911).

A construction contractor by trade, Frank Gilbreth gave up that career to study scientific management after hearing Taylor speak at a professional meeting. Frank and his wife, Lillian, a psychologist, focused their studies on eliminating inefficient hand-and-body motions. The Gilbreths also experimented with the design and use of the proper tools and equipment for optimizing work performance.[4]

Frank is probably best known for his bricklaying experiments. By carefully analyzing the bricklayer's job, he reduced the number of motions in laying exterior brick from eighteen to about five, and in laying interior brick from eighteen to two. Using Gilbreth's techniques, a bricklayer was far more productive and less fatigued at the end of the day.

The Gilbreths were among he first researchers to use motion pictures to study hand-and-body motions. They invented a device called a microchronometer that recorded a worker's hand-and-body motions and the amount of time spent doing each motion. Wasted motions missed by the naked eye could be identified and eliminated. The Gilbreths also devised a classification scheme to label seventeen basic hand motions (such as search, grasp, hold), which they called **therbligs** ("Gilbreth" spelled backward, with the *th* transposed). This scheme gave the Gilbreths a more precise way of analyzing a worker's exact hand movements.

therbligs
A classification scheme for labeling basic hand motions

General administrative theory
An approach to management that focuses on describing what managers do and what constitutes good management practice

principles of management
Fundamental rules of management that could be applied in all organizational situations and taught in schools

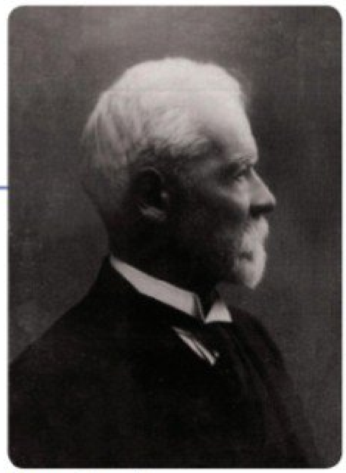
bureaucracy
A form of organization characterized by division of labor, a clearly defined hierarchy, detailed rules and regulations, and impersonal relationships

HOW TODAY'S MANAGERS USE SCIENTIFIC MANAGEMENT As noted earlier at UPS, many of the guidelines and techniques Taylor and the Gilbreths devised for improving production efficiency are still used in organizations today. When managers analyze the basic work tasks that must be performed, use time-and-motion study to eliminate wasted motions, hire the best-qualified workers for a job, or design incentive systems based on output, they're using the principles of scientific management.

General Administrative Theory

General administrative theory focused more on what managers do and what defined good management practice.

We introduced Henri Fayol in Chapter 1 because he first identified five functions that managers perform: planning, organizing, commanding, coordinating, and controlling.[5] Fayol wrote during the same time period as Taylor. While Taylor was concerned with first-line managers and the scientific method, Fayol's attention was directed at the activities of *all* managers. He wrote from his personal experience as the managing director of a large French coal-mining firm.



Henry Fayol's
Henry Fayol's general administrative theory focused on what managers do and what defined good management practice.
Source: Historic Collection/Alamy Stock Photo

Fayol described the practice of management as something distinct from accounting, finance, production, distribution, and other typical business functions. His belief that management was an activity common to all business endeavors, government, and even the home led him to develop fourteen **principles of management**—fundamental rules of management that could be applied to all organizational situations and taught in schools. These principles are shown in Exhibit MH-3.

Max Weber (pronounced VAY-ber) was a German sociologist who studied organizations.[6] Writing in the early 1900s, he developed a theory of authority structures and relations based on an ideal type of organization he called a **bureaucracy**—a form of organization characterized by division of labor, a clearly defined hierarchy, detailed

Exhibit MH-3

Fayol's Fourteen Principles of Management

1. **Division of work.** Specialization increases output by making employees more efficient.
2. **Authority.** Managers must be able to give orders, and authority gives them this right.
3. **Discipline.** Employees must obey and respect the rules that govern the organization.
4. **Unity of command.** Every employee should receive orders from only one superior.
5. **Unity of direction.** The organization should have a single plan of action to guide managers and workers.
6. **Subordination of individual interests to the general interest.** The interests of any one employee or group of employees should not take precedence over the interests of the organization as a whole.
7. **Remuneration.** Workers must be paid a fair wage for their services.
8. **Centralization.** This term refers to the degree to which subordinates are involved in decision-making.
9. **Scalar chain.** The line of authority from top management to the lowest ranks is the scalar chain.
10. **Order.** People and materials should be in the right place at the right time.
11. **Equity.** Managers should be kind and fair to their subordinates.
12. **Stability of tenure of personnel.** Management should provide orderly personnel planning and ensure that replacements are available to fill vacancies.
13. **Initiative.** Employees allowed to originate and carry out plans will exert high levels of effort.
14. **Esprit de corps.** Promoting team spirit will build harmony and unity within the organization.

Source: Based on Henri Fayol's 1916 "Principles of Management," in *Administration Industrielle et Générale,* translated by C. Storrs as *General and Industrial Management* (London: Sir Isaac Pitman & Sons, 1949).

Max Weber

Max Weber developed a theory of authority structures and relations based on an ideal type of organization he called a bureaucracy.

Source: Keystone Pictures USA/ZumaPress/Alamy Stock Photo

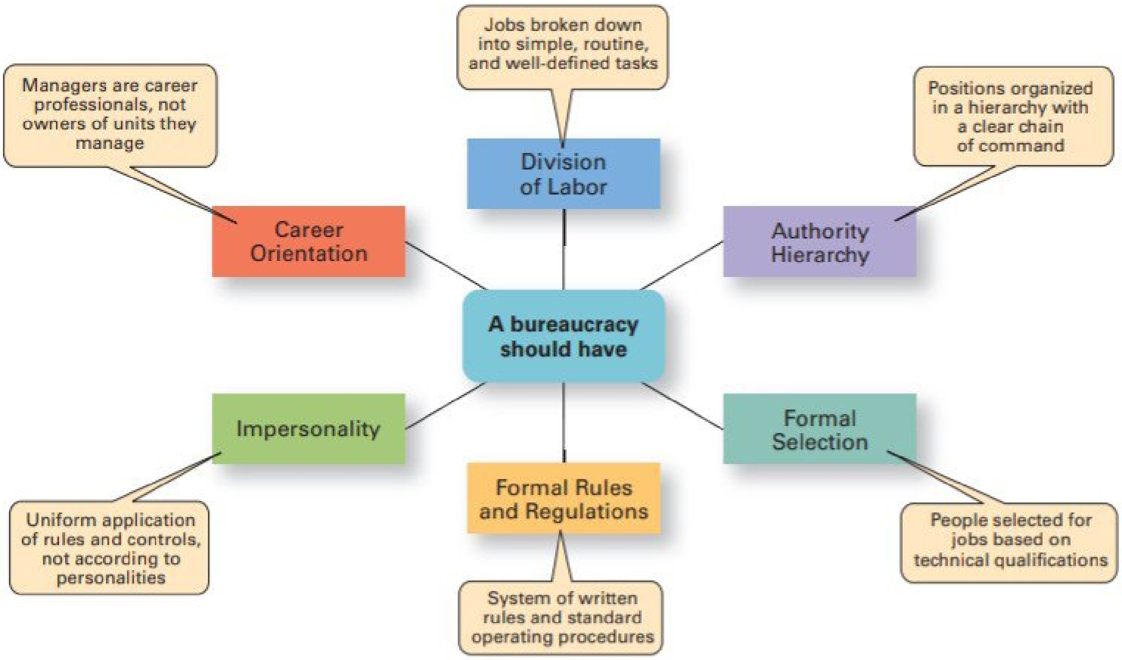
rules and regulations, and impersonal relationships. (See Exhibit MH-4.) Weber recognized that this "ideal bureaucracy" didn't exist in reality. Instead, he intended it as a basis for theorizing about how work could be done in large groups. His theory became the structural design for many of today's large organizations.

Bureaucracy, as described by Weber, is a lot like scientific management in its ideology. Both emphasized rationality, predictability, impersonality, technical competence, and authoritarianism. Although Weber's ideas were less practical than Taylor's, the fact that his "ideal type" still describes many contemporary organizations attests to their importance.

HOW TODAY'S MANAGERS USE GENERAL ADMINISTRATIVE THEORY Several of our current management ideas and practices can be directly traced to the contributions of general administrative theory. For instance, the functional view of the manager's job can be attributed to Fayol. In addition, his fourteen principles serve as a frame of reference from which many current management concepts—such as managerial authority, centralized decision making, reporting to only one boss, and so forth—have evolved.

Weber's bureaucracy was an attempt to formulate an ideal prototype for organizations. Although many characteristics of Weber's bureaucracy are still evident in large organizations, his model isn't as popular today as it was in the twentieth century. Many managers feel that a bureaucratic structure hinders individual employees' creativity and limits an organization's ability to respond quickly to an increasingly dynamic environment. However, even in flexible organizations of creative professionals—such as Apple, Samsung, or Cisco Systems—bureaucratic mechanisms are necessary to ensure that resources are used efficiently and effectively.[7]

Exhibit MH-4
Characteristics of Weber's Bureaucracy



Source: Based on *Essays in Sociology* by Max Weber, translated, edited, and introduced by H. H. Gerth and C. Wright Mills (New York: Oxford University Press, 1946).